

An
Internship Report
On



Machine Learning with AI Using Python
Along with the Project
INTELLIGENT SENTIMENT DETECTION
SYSTEM

Submitted in partial fulfilment of the award of the degree of
UNDER GRADUATION

In
COMPUTER APPLICATION

Submitted by
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UNDER THE GUIDANCE OF
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SUBMITTED TO
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DEPARTMENT OF COMPUTER SCIENCE

KSAWUV, MANDYA

CERTIFICATE

This is to certify that the project entitled “INTELLIGENT SENTIMENT DETECTION SYSTEM FOR ONLINE CUSTOMER REVIEW” is a project work carried out by S R LEKHANA [U10UD22S0002] in partial fulfilment of the requirement for the award of the degree of BACHELOR OF COMPUTER APPLICATION from the KSAWUV, MANDYA is based on the result of the project work carried out during the year 2024-25. This project or any part thereof has not been submitted for any purpose to any other University or institute.

Signature of the Guide

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Name & Signature of the Examiners

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**Department of Women's Studies
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MACHINE LEARNING WITH AI USING PYTHON

Company Name: Jupiterking Technologies

Role: Machine Learning Intern

Duration: 4 Weeks

❖ PROFILE OF THE ORGANIZATION

Name of the organization: Jupiterking Technologies pvt.Ltd.

Address: #MIG 20, Kuvempunagara, Mysore -570023.

Mission & Vision: Jupiterking technologies specializes in developing and integrating Machine learning technologies, Web development, Full stack development.

Provide an opportunity to work on company projects and mentorship from experienced employees of Jupiterking Technologies to enhance your technical skills.

❖ TEAM OVERVIEW

The team consists of 5 members, including interns, project managers, research professionals working together to solve problems using data-driven techniques.

A Machine Learning (ML) team in an internship setting typically focuses on developing, testing, and deploying ML models for real-world applications.

DOMAIN INTRODUCTION

Machine learning is a branch of Artificial Intelligence that focuses on developing models and algorithms that let computers learn from data without being explicitly programmed for every task. In simple words, ML teaches the systems to think and understand like humans by learning from the data.

It enables computers to identify patterns, make decisions, and improve performance over time with minimal human intervention. Machine learning is widely used in various applications such as image recognition, natural language processing, recommendation systems, and autonomous vehicles. By continuously analysing new data, ML models adapt and become more accurate, making them essential for solving complex real-world problems. The growing

availability of big data and advances in computational power have accelerated the development and adoption of machine learning technologies across industries.

TYPES OF MACHINE LEARNING

Machine Learning is generally divided into three main types:

1. Supervised Learning

- In supervised learning, the model is trained on a labeled dataset, meaning each training example is paired with the correct output.
- The goal is for the model to learn the mapping between inputs and outputs.
- It is used for **classification** (e.g., spam detection) and **regression** (e.g., predicting house prices).
- **Example algorithms:** Linear Regression, Logistic Regression, Decision Trees, Support Vector Machines (SVM).

2. Unsupervised Learning

- In unsupervised learning, the data has no labels; the algorithm tries to find hidden patterns or structures in the data.
- It is commonly used for **clustering** and **dimensionality reduction**.
- **Example use case:** grouping customers by purchasing behavior.
- **Example algorithms:** K-Means Clustering, Hierarchical Clustering, PCA (Principal Component Analysis).

3. Reinforcement Learning

- In reinforcement learning, an agent learns to make decisions by interacting with an environment.
 - It receives **rewards or penalties** based on its actions and learns a strategy to maximize the total reward.
 - Often used in robotics, game AI, and self-driving cars.
 - **Example algorithms:** Q-Learning, Deep Q-Networks (DQN), SARSA.
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Advantages of Machine Learning

1. Automation of Tasks

ML automates repetitive and time-consuming tasks, reducing human effort.
Once trained, models can perform tasks continuously with minimal supervision.

2. Improved Decision Making

ML analyses large amounts of data to help make data-driven decisions.
It can uncover insights that are difficult for humans to identify manually.

3. Efficient Handling of Large Data

ML can process and analyze massive datasets quickly and accurately.
It helps organizations leverage big data for better outcomes.

4. Wide Range of Applications

ML is used in various fields such as healthcare, finance, agriculture, and e-commerce.
Its flexibility allows it to solve diverse real-world problems.

5. Real-Time Predictions

ML systems can provide instant results based on live input data.
This is useful in areas like fraud detection, navigation, and online recommendations.

6. Personalization

ML helps tailor content and services to individual users (e.g., Netflix, Amazon).
It enhances customer experience through targeted suggestions.

Applications of Machine Learning

1. Image and Video Recognition

ML is used in facial recognition, object detection, and medical image analysis.
Examples include face unlock on phones and detecting tumours in X-rays.

2. Natural Language Processing (NLP)

Used for language translation, sentiment analysis, and chatbots.
ML helps machines understand, generate, and respond to human language.

3. Fraud Detection

In banking and finance, ML detects unusual transaction patterns.
It helps prevent credit card fraud and other financial crimes in real time.

4. Recommendation Systems

ML provides personalized recommendations on platforms like Netflix, YouTube, and Amazon.

It analyses user behaviour to suggest relevant content or products.

5. Autonomous Vehicles

Self-driving cars use ML to recognize roads, obstacles, and traffic signs.

ML helps in decision-making for navigation and safety.

6. Healthcare and Diagnosis

ML assists in early disease detection, drug discovery, and treatment planning.

It improves diagnosis accuracy using medical data and imaging.

7. Customer Support Automation

Chatbots and virtual assistants use ML to handle customer queries.

They provide fast, consistent, and 24/7 support services.

8. Predictive Maintenance

ML predicts equipment failures in industries before they occur.

This reduces downtime and saves maintenance costs.

Intelligent Sentiment Detection System for Online Customer Reviews – Amazon Product Reviews.

1. INTRODUCTION OF THE PROJECT

1.1 Overview of the project

An Intelligent Sentiment Detection System is designed to automatically analyze and interpret the sentiment expressed in a piece of text, such as product reviews, social media posts, or customer feedback. The main goal of such a system is to classify text into predefined sentiment categories, typically positive, negative, or neutral, but sometimes extended to more granular categories, such as very positive, very negative, or mixed sentiments.

1.2 Objectives:

- To develop a sentiment classification model using Naïve Bayes.
- To preprocess and vectorize text data for feature extraction.
- To evaluate the performance of the model using accuracy and classification metrics.

2. TOOLS AND TECHNOLOGIES

Back-end development tool: PyCharm

2.1 Core Libraries for Data Handling

- **NumPy** – For numerical computing and handling arrays.
- **Pandas** – For data manipulation and analysis.
- **Matplotlib & Seaborn** – For data visualization, including the confusion matrix heatmap.

2.2 Machine Learning Libraries

- **Scikit-learn** – A fundamental library for ML algorithms (classification, regression, clustering).
- **zip file** - For extracting the dataset from a compressed format.

2.3 Natural Language Processing (NLP) Tools

- **NLTK** - For NLP tasks like tokenization and sentiment analysis.
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3. SETTING UP TOOLS AND TECHNOLOGIES

3.1 Download & Install Python

Download Python: Visit the official Python website:

<https://www.python.org/downloads>

Install Python: Open the installer and follow the on-screen instructions.

Verify Installation: After installation, verify that Python is correctly installed opening a command prompt or terminal and typing: [python --version](#)

3.2 Download & Install PyCharm

- i. Go to the [JetBrains website](#).
- ii. Download **PyCharm Community Edition (Free)** or **Professional Edition (Paid)**.
- iii. Install it following the on-screen instructions.

3.2 Configure Python Interpreter

Set Up a Python Virtual Environment:

It is highly recommended to set up a virtual environment for your project to avoid conflicts with other Python packages on your system.

- i. Open **PyCharm** and create a new project.
- ii. Click **File** → **Settings** → **Project: <Project Name>** → **Python Interpreter**.

3.3 Install Required Libraries

To import this package, write a command in terminal -

[pip install <package name>](#)

numpy, pandas, matplotlib, seaborn, scikit-learn

3.4 Run Code

Click the Run button in PyCharm (green triangle) to execute your text detection program and check the results.

4. METHODOLOGY [TASKS & RESPONSIBILITIES]

4.1 Algorithm/Model

Naive Bayes: A probabilistic classifier based on Baye's theorem (to predict probability of data), assuming independence between features.

It works well when features (e.g., words) are conditionally independent, making it suitable for text classification.

Types - Multinomial NB

- Bernoulli NB
- Gaussian NB.

TF-IDF (Term Frequency-Inverse Document Frequency): A feature extraction technique used to transform text data into numerical values, emphasizing important words in the dataset.

4.1 Data Collection

The dataset used in this project is Amazon product reviews from the sentiment-labelled sentences dataset. Each review was preprocessed and converted numerical format suitable for machine learning models.

The dataset consists of **1,000 reviews** with labels:

- 0 → Negative review
- 1 → Positive review

Each record contains: **Review Text:** The customer's feedback on the product.

Sentiment Label: The sentiment classification (0 or 1).

4.2 Data Preprocessing

Textual data is cleaned and preprocessed, including steps like tokenization, removing stop words, lemmatization, and stemming.

- i. Downloaded and loaded the dataset.
 - ii. Extracted raw text and corresponding sentiment labels.
 - iii. Split the data into training (80%) and testing (20%) sets.
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- iv. Applied **TF-IDF Vectorization** to convert text data into numerical features, removing stopwords (common words like ‘the’, ‘is’, ‘and’) and limiting vocabulary size to improve the accuracy

4.3 Feature Extraction

- The text reviews were transformed into numerical data using **Term Frequency-Inverse Document Frequency (TF-IDF)**.
- This method gives higher weights to unique words that appear less frequently across all reviews.

4.4 Sentiment Analysis Models

- The Multinomial Naïve Bayes (MultinomialNB) algorithm was used for sentiment classification.
- It is well-suited for text classification as it assumes word probabilities are independent, making it efficient.

4.5 Visualization and Reporting

Once the sentiment is detected, it is visualized through dashboards, graphs, map or reports to help users quickly grasp insights into customer feelings, trends, or market responses.

4.6 Evaluation Metrics

To compare the models, the following metrics were used:

- **Accuracy:** Percentage of correctly classified reviews.
- **Classification Report:** Precision, recall, and F1-score for positive and negative sentiment classifications.

5. RESULTS AND ANALYSIS

5.1 Model Performance - The performance of the model was assessed using:

- **Accuracy Score:** Measures the percentage of correctly classified reviews.
 - **Precision, Recall, and F1-Score:** Provides deeper insights into the model’s performance.
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METRIC	SCORE
Accuracy	77%
Precision (Negative)	81%
Precision (Positive)	74%
Recall (Negative)	70%
Recall (Positive)	84%
F1-Score (Negative)	75%
F1-Score (Positive)	79%

5.1 Key Findings

- The Naïve Bayes classifier achieved an **77% accuracy** in classifying Amazon product reviews.
- Misclassification mainly occurred in ambiguous or sarcastic reviews.
- TF-IDF performed well in extracting important features from the text.

5.2 Visualization

A confusion matrix heatmap was plotted using seaborn to visualize the model's predictions, showing the number of correct and incorrect classifications.

- This helps visualize how well the classifier predicted positive and negative sentiments.
- The x-axis represents **Predicted Labels** (Negative, Positive).
- The y-axis represents **Actual Labels** (Negative, Positive).
- The colour intensity indicates the count of predictions in each category.

5.3 Insights & Observations

- The model performed well in classifying positive and negative reviews with a reasonable accuracy.
 - The model's performance could be improved further by using deep learning-based NLP techniques like LSTMs or transformers.
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- The dataset's size and balance could impact performance, and using a larger dataset may yield better results.

6. CHALLENGES FACED

- **Handling Ambiguities:** Handling imbalanced datasets where positive reviews outnumbered negative ones.

Some sentences may carry mixed emotions, and accurately classifying them can be challenging for basic sentiment analysis models.

- **Text Preprocessing:** Dealing with sarcasm and ambiguous text in reviews.

For a real-world application, preprocessing steps like tokenization, stop word removal, and lemmatization would be necessary for better accuracy.

- Computational limitations while training deep learning models.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project successfully implemented Naïve Bayes for sentiment analysis of Amazon product reviews.

The model effectively classified positive and negative reviews, achieving a high accuracy rate.

7.2 Future Scope

- **Improving Accuracy:** Experimenting with Logistic Regression, SVM, or Deep Learning (LSTMs).
 - **Handling Neutral Sentiments:** Adding a third category for neutral reviews.
 - **Expanding Dataset:** Using a larger dataset for better generalization.
 - **Aspect-Based Sentiment Analysis:** Identifying sentiment towards specific product features.
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8. REFERENCES

- scikit-learn documentation: <https://scikit-learn.org>
- Sentiment Analysis Concepts: <https://towardsdatascience.com/sentiment-analysis>

9. GRATITUDE

I would like to extend my gratitude to my mentor, Mrs.Akshatha, Mrs.Nithya Ram and the entire development team at Jupiterking Technologies Solutions for their invaluable guidance and support throughout my internship.

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